

SMARTGYM DISPLAY BRING-UP

This project is intentionally separate from 'SmartGym'.

Purpose

Use this repo-sized sandbox to validate the new VIEWS 7-inch ESP32-S3 board:

- upload flow
- serial over USB
- flash / PSRAM assumptions
- panel initialization
- GT911 touch
- LVGL rendering

Why Separate

The new display board changes too many things at once:

- ESP32 -> ESP32-S3
- GPIO map
- display stack
- touch stack
- USB/serial behavior
- memory profile

Keeping bring-up separate makes debugging much easier and avoids contaminating the working SmartGym MVP while hardware setup is still unstable.

Migration Strategy

1. Get this project stable with basic panel/touch/LVGL tests.
2. Keep SmartGym unchanged while testing.
3. Once stable, migrate the display knowledge back into SmartGym as a proper HMI layer.
4. Reuse existing SmartGym domain modules later:
 - 'AppController'
 - 'RepDetector'
 - 'CalibrationService'
 - 'SessionRecorder'
 - 'FirebaseService'
 - 'UserRegistry'
 - 'MachineRegistry'

Next Implementation Step

After confirming this project flashes and reports PSRAM correctly, the next task should be a minimal panel-on + touch test screen.

Cloud Reconciliation Policy (Current Firmware)

When an RFID card is scanned and cloud is enabled, the device runs a reconciliation pass before training:

1. Pull latest user profile ('usersByRfid/{uid}').
2. Pull latest calibration ('calibrations/{uid}/{machineTypeId}').
3. Pull latest machine config ('machineConfigs/{machineId}') during poll cycle.
4. Merge conflict rule:
 - If cloud timestamp is newer, local profile/calibration is updated.
 - If local timestamp is newer, local data is pushed back to cloud.
5. Session uploads are idempotent:
 - deterministic paths by 'sessionId' and timeline key
 - payloads include 'schemaVersion' and 'idempotencyKey'.